

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 10: Circles

Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board paper and the marks. Solutions are in a separate PDF with the same question numbers.

1 Theorem 1: tangent and radius

- Assertion (A): The tangents drawn at the end points of a diameter of a circle are parallel.
Reason (R): Diameter of a circle is the longest chord. Choose: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. **CBSE 2024, 1 mark**

2 Theorem 2: tangents from an external point

- AB is a diameter of a circle with centre O . AQ , BP and PQ are tangents to the circle, where Q lies on the tangent at A and P lies on the tangent at B . Prove that $\angle POQ = 90^\circ$. **CBSE 2024, 3 marks**
- A circle with centre O and radius 8 cm is inscribed in a quadrilateral $ABCD$ in which P , Q , R , S are the points of contact on AD , DC , CB and BA respectively. If AD is perpendicular to DC , $BC = 30$ cm and $BS = 24$ cm, then find the length DC . **CBSE 2024, 3 marks**
- A backyard is in the shape of a triangle ABC with right angle at B . $AB = 7$ m and $BC = 15$ m. A circular pit was dug inside it such that it touches the walls AC , BC and AB at P , Q and R respectively such that $AP = x$ m. (i) Find the length of AR in terms of x . (ii) Write the type of quadrilateral $BQOR$. (iii) Find the length PC in terms of x and hence find the value of x , OR find x and hence find the radius r of the circle. **CBSE 2024, 4 marks, case study**

